

Business Context

The used-car resale industry is becoming increasingly data-driven, with organizations handling large volumes of historical vehicle market data to support pricing, inventory, and market analysis decisions. The need for quick access to reliable insights from this data is critical for responding to market changes, optimizing inventory, and maintaining competitive pricing in a fast-paced environment.

Data teams often face challenges in efficiently extracting insights from complex datasets that include multiple vehicle attributes such as model, year, mileage, fuel type, and transmission. Translating business questions into accurate data analysis code can be time-consuming, particularly when similar exploratory analyses are performed repeatedly. This challenge is amplified by growing expectations for faster turnaround times and consistent analytical outputs.

To address these challenges, organizations can focus on adopting solutions that simplify data interaction and accelerate insight generation. Enabling natural language driven analysis allows data analyst teams to access data more intuitively, reduces manual effort, and supports timely, data-backed decision-making across business teams.

Objective

The objective is to simplify and streamline the day-to-day work of data analysts by enabling natural language interaction with used-car market data. Common analytical requests can be translated into executable data analysis code, reducing the time spent on writing, debugging, and repeating similar logic. This allows data analysts to focus more on analysis and validation rather than manual coding.

- Reduce time spent on repetitive data analysis tasks
- Improve consistency and reliability of analysis code
- Support faster turnaround for routine analytical requests

Data Description

The dataset consists of several key attributes describing vehicle listings:

- **ID:** Unique identifier assigned to each vehicle listing.
- **Region:** Categorical variable representing the broader geographic region where the vehicle is listed.
- **Price:** Numerical value indicating the listed price of the vehicle.
- **Year:** Numerical value representing the model year of the vehicle.
- **Manufacturer:** Categorical variable identifying the vehicle's manufacturer.
- **Model:** Categorical variable specifying the model name of the vehicle.
- **Cylinders:** Categorical variable describing the engine cylinder configuration (for example, 4 cylinders, 6 cylinders).

Problem Statement: CodeGen Analyst Agent

year, mileage, color, etc.

- **Transmission:** Categorical variable specifying the type of transmission (automatic or manual).
- **Drive:** Categorical variable describing the drivetrain configuration (for example, front-wheel drive, rear-wheel drive, all-wheel drive).
- **Type:** Categorical variable indicating the vehicle body type (such as sedan, SUV, truck, or coupe).
- **Paint Color:** Categorical variable representing the exterior color of the vehicle.
- **Description:** Textual field containing the seller-provided description of the vehicle.
- **State:** Categorical variable identifying the state in which the vehicle is listed.
- **Latitude:** Numerical value representing the latitude of the listing location.
- **Longitude:** Numerical value representing the longitude of the listing location.
- **Posting Date:** Date field indicating when the vehicle listing was posted.

This dataset captures a wide variety of vehicle characteristics, pricing information, and geographic details, enabling comprehensive analysis of used car markets and supporting tasks such as price prediction, demand analysis, and regional trend identification.

